

Notes: Barlevy (RES, 2002)

The Sullyng Effect of Recessions

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1 Big picture

Barlevy (2002) revisits the Schumpeterian view that recessions are useful because they “cleanse” the economy of bad jobs and reallocations. In the standard cleansing story, a recession lowers profitability, marginal low-productivity matches are destroyed, and the average productivity of surviving jobs rises. The paper accepts that this force exists, but argues that it is incomplete once workers can search while employed.

The central idea is the **sullyng effect**. Recessions reduce vacancy creation. When vacancies are scarce, employed workers have fewer chances to move from mediocre jobs to better jobs. Therefore, many workers remain stuck in less productive matches. The economy may destroy the worst jobs, but it also produces fewer transitions into the best jobs. The net allocation can become worse, not better.

The paper’s calibration suggests that the sullyng effect is likely to dominate the cleansing effect. In the benchmark simulation, a recession initially raises average match quality slightly because the worst jobs are destroyed, but after several quarters average match quality falls below its prerecession level. In a simulated path calibrated to U.S. unemployment from the 1970s and early 1980s, cumulative recessions reduce average match productivity by almost 1 percent.

2 Incremental contribution of the paper

1. **Adds on-the-job search to the cleansing literature.** Previous models emphasized destruction of low-productivity jobs. Barlevy shows that job-to-job mobility is equally important for understanding the allocation of workers across matches.
2. **Separates two opposing forces.** Recessions have a cleansing effect through higher job-destruction thresholds, but also a sullyng effect through reduced vacancy creation and weaker upward mobility.
3. **Explains procyclical job quality.** The paper provides a theory consistent with evidence that jobs created in recessions tend to be lower-paying, more temporary, or more likely to be destroyed.
4. **Connects quits, mismatch, and productivity.** The model links the procyclicality of quits and job-to-job flows to the countercyclicality of mismatch and underutilization.
5. **Quantifies the mechanism.** The calibrated model uses unemployment and quit-rate behavior to discipline the magnitude of job-finding probabilities and then studies the implied movement in match quality.
6. **Changes the interpretation of reallocation.** More job destruction in recessions does not automatically mean more efficient resource allocation, because the distribution of surviving matches may deteriorate.

3 Main research question

The paper asks:

Do recessions improve the allocation of labor by destroying bad matches, or do they worsen allocation by trapping workers in mediocre jobs?

The question matters because the empirical evidence is mixed. Job destruction and job reallocation rise in downturns, which seems consistent with cleansing. But worker quits and job-to-job moves are procyclical, and jobs created in recessions often look lower quality. Barlevy argues that these facts are not contradictory: the same recession can destroy the worst matches while slowing the movement of workers toward the best matches.

4 Conceptual mechanism

4.1 Cleansing

The cleansing effect is the traditional mechanism. A negative aggregate productivity shock lowers the surplus generated by every match. Matches that were barely viable in good times become unprofitable and are destroyed. Because the least productive surviving matches disappear, the average quality of remaining matches mechanically rises.

This force is present in the model. There is a cutoff match quality $n^*(y)$ such that matches below the cutoff are not viable. Since lower aggregate productivity reduces match surplus, the cutoff rises in recessions. This is the same logic that appears in Mortensen and Pissarides-type models with endogenous job destruction.

4.2 Sullyng

The sullyng effect works through the creation side. During recessions, lower expected surplus reduces vacancy posting. With fewer vacancies per searcher, workers have fewer chances to encounter better jobs. Since employed workers search on the job, reduced vacancy creation slows their upward movement across match qualities.

This matters because the stock of workers in the economy is not only divided into employed and unemployed. Employed workers are also distributed across match qualities. If job-to-job mobility slows, fewer workers reach high-quality matches before exogenous separation sends them back into unemployment. The mass of workers shifts toward mediocre matches.

4.3 Net effect

The net effect of a recession on average match quality is ambiguous in theory:

- cleansing removes the lowest-quality matches and raises average match quality;
- sullyng reduces upward job-to-job mobility and lowers average match quality among surviving jobs.

The quantitative question is which force is larger. Barlevy's calibration finds that cleansing produces a small short-run improvement, while sullyng creates a larger and more persistent deterioration.

5 Model environment

5.1 Workers, jobs, and match quality

There are N worker types and N job types. A worker of type i can be employed in any job of type j , but productivity depends on the distance between i and j . The match quality index is denoted $n \in \{1, \dots, N\}$, where higher n means a better match. The best match is $n = N$.

Output in a match is

$$y_{ij} = y(1 + \epsilon_{ij}),$$

where y is aggregate productivity and ϵ_{ij} is match-specific productivity. The values are ordered so that

$$\epsilon_1 < \epsilon_2 < \dots < \epsilon_N.$$

Workers can also produce b units outside a firm. This outside option pins down the idea that a job should only be maintained if it creates enough surplus relative to nonemployment.

5.2 Search and matching

Entrepreneurs post vacancies at cost k . Workers who are unemployed and workers who are employed in less-than-best matches can search. This is the key departure from simpler cleansing models: workers do not have to become unemployed to search for a better match.

Let u denote unemployment and e_n the fraction of workers employed in match quality n . The measure of searchers is

$$z = u + \sum_{n=1}^{N-1} e_n.$$

Only workers in the best possible match $n = N$ do not search. Total vacancies are v , and contacts are generated by a constant-returns matching function $m(z, v)$.

A worker accepts a new job only if it is better than the current match. Thus an unemployed worker may accept any viable match, while a worker already in a medium-quality job accepts only a higher-quality job. This generates gradual upward mobility through job-to-job transitions.

5.3 Vacancy creation and free entry

Free entry drives the expected value of posting a vacancy to zero. The expected value of a vacancy depends on three forces:

1. the probability that the vacancy meets a worker;
2. the quality of the match if a worker is found;
3. the probability that the worker accepts the offer, which is higher if many workers are trapped in low-quality matches.

When aggregate productivity falls, the surplus from any match falls. To restore zero expected profits, vacancies must become scarcer relative to searchers. This is why the vacancy-searcher ratio v/z falls in recessions, making job-to-job mobility harder.

5.4 Wage setting

In the steady-state model, wages split match surplus between the worker and the entrepreneur. If Π_n is the entrepreneur's value, W_n is the worker's value, U is the value of unemployment, and V is the value of a vacancy, the surplus is

$$S_n = \Pi_n + W_n - U - V.$$

The worker receives share α of the surplus:

$$W_n - U = \alpha S_n.$$

For the numerical model, the paper simplifies by assuming that worker and entrepreneur split output rather than surplus. This makes the computation more tractable because the value functions can be solved separately by match quality.

6 Steady-state results

6.1 Proposition 1: unique symmetric steady state

With constant aggregate productivity, there is a unique symmetric steady-state equilibrium. Symmetry means that the number of vacancies and the distribution of workers across match qualities do not depend on the label of the worker or job type. This result establishes a well-defined benchmark from which comparative statics can be studied.

The steady state still contains mismatch. Unless only the best match is viable, some workers remain in non-best jobs because posting vacancies is costly. Search frictions prevent instant assignment of each worker to the job where the worker is most productive.

6.2 Proposition 2: the cleansing effect

There exists a cutoff match quality $n^*(y)$ such that a match is maintained if and only if $n \geq n^*(y)$. The cutoff is weakly decreasing in aggregate productivity y . Equivalently, when y falls in a recession, the cutoff rises and low-quality jobs are destroyed.

This is the cleansing effect. If the only allocation margin were destruction, average match quality would rise in recessions. This is the logic behind earlier cleansing models.

6.3 Proposition 3: the sullyng effect

Consider two steady states with $y_1 > y_0$. The vacancy-searcher ratio v/z is higher in the high-productivity steady state. Moreover, conditional on matches surviving the low-productivity cutoff, the distribution of workers across match qualities under y_1 first-order stochastically dominates the distribution under y_0 .

The implication is direct: recessions lower the arrival rate of better job offers. Workers move up the match-quality ladder more slowly, so fewer workers end up in the best jobs. This is the formal sullyng result.

7 Calibration and quantitative exercise

7.1 Why simulation is needed

Once aggregate productivity fluctuates, the entire distribution $\{e_n\}_{n=1}^N$ becomes a state variable. This makes transitional dynamics difficult to solve analytically. The paper therefore solves a discrete-time version of the model numerically.

The simulation has two purposes. First, it verifies that the steady-state intuition survives when shocks are temporary and agents are forward-looking. Second, it asks whether the sully effect is quantitatively large enough to matter.

7.2 Key calibration choices

The benchmark model uses quarterly periods. Aggregate productivity takes two values, $y_1 = 1$ and $y_0 < 1$, with a probability of switching equal to 0.05 per quarter. A productivity state therefore lasts about 5 years on average, matching business-cycle frequencies.

The model sets $N = 6$ match qualities. This is motivated partly by Topel and Ward's evidence that young workers move through roughly 6 to 7 jobs during their first 10 years, suggesting multiple steps in the match-quality ladder.

The discrete-time matching function is

$$m(z, v) = \frac{zv}{z + v},$$

which is constant returns to scale and bounded by the smaller side of the market.

The output split is set at $\alpha = 0.5$, and the quarterly discount rate is $r = 0.01$. Match-specific productivity values are chosen so that a worker expects about a 10 percent wage gain when moving to a better match, consistent with evidence from job mobility studies. The paper reports the sequence

$$\begin{aligned}\epsilon_1 &= 0.0167, & \epsilon_2 &= 0.0367, & \epsilon_3 &= 0.0617, \\ \epsilon_4 &= 0.0950, & \epsilon_5 &= 0.1450, & \epsilon_6 &= 0.2450.\end{aligned}$$

The low aggregate productivity value, vacancy cost, and exogenous separation rate are calibrated to match key unemployment moments. The paper reports

$$y_0 = 0.280, \quad k = 1.892, \quad s = 0.018.$$

The calibration makes long-run unemployment fluctuate between roughly 3 percent and 9 percent, and limits the immediate jump in unemployment after a recession shock to a realistic size.

7.3 Main quantitative findings

The simulation yields five important findings:

1. A recession sharply lowers the probability of finding a job from about 0.36 to 0.13 on impact, with partial recovery to about 0.18.

2. The model produces procyclical quits, consistent with the PSID evidence.
3. The share of mismatched workers rises in recessions, both in the data and in the model.
4. Average match productivity initially rises slightly because of cleansing, but then falls below its prerecession level as sullyng accumulates.
5. The quality of newly created matches can be lower during recessions after the initial cleansing spike.

The most important quantitative conclusion is that the sullyng effect is larger and more persistent than the cleansing effect. A recession can look cleansing at first, but its cumulative effect on the allocation of workers may be negative.

8 Empirical discipline and interpretation

8.1 Quit rates

The paper uses quit rates from the Panel Survey of Income Dynamics as a check on the model. Quits are important because job-to-job mobility is the channel through which workers improve match quality. In the data, quit rates are procyclical: workers quit more in booms and less in recessions. This supports the mechanism that recessions reduce the movement of workers into better jobs.

The model can reproduce this qualitative procyclical, although it produces too much high-frequency quit volatility. The paper interprets the long-run behavior as more reliable than the high-frequency spikes.

8.2 Mismatch

Mismatch is measured in the model as the fraction of employed workers not in their best match, $n < N$. In the PSID, the paper uses workers' answers to whether they have thought about changing jobs as a suggestive empirical analogue.

Both the data and the model imply countercyclical mismatch. The model predicts that after the recessions between 1970 and 1983, roughly 8 additional percentage points of workers shift into jobs where they are mismatched. The paper stresses that the exact level is hard to compare, because the survey question is subjective, but the cyclical, is informative.

8.3 Average match productivity

Average match productivity summarizes both cleansing and sullyng. It is the average of $1 + \epsilon_n$ across existing matches. The simulation shows that the initial cleansing effect is small: average match quality rises from about 1.219 to 1.221. But after several quarters, the sullyng effect dominates. After 5 years, average match productivity falls to about 1.211 and converges in the long run to about 1.206.

The bottom panel of Figure 5 applies the model to a path that mimics U.S. unemployment. It

predicts a cumulative decline of almost 1 percent in average match productivity by the peak unemployment period. This is the main quantitative challenge to the pure cleansing view.

9 What to remember

The paper’s core lesson is that **job destruction is not the same thing as efficient reallocation**. Recessions may destroy low-quality jobs, but they also weaken the job-finding and job-to-job mobility process that moves workers into high-quality jobs. A recession therefore has two allocation effects, and the second may dominate.

The model also changes how to interpret procyclical quits. In many models, fewer quits in recessions simply mean fewer separations. Here, fewer quits mean less movement up the match-quality ladder. This creates a stock of workers in mediocre jobs and lowers future productivity.

10 Limitations and caveats

- **Calibration requires a large productivity shock.** The paper notes that the implied drop in aggregate productivity is implausibly large. The author argues that this matters less for the sullyng calculation because the key object is the job-finding probability, which can be disciplined by unemployment dynamics.
- **The model is stylized.** Workers and jobs live on a symmetric match-quality circle; this is useful for tractability but not meant as a literal description of occupations or firms.
- **No endogenous search intensity in the main simulation.** Workers do not choose how hard to search. The paper discusses that endogenous search could amplify or dampen sullyng, depending on the cost of search and the return to better matches.
- **Survey mismatch is imperfect.** The PSID question about wanting a new job is only a rough proxy for the model’s notion of mismatch.
- **The flow results are sensitive.** New-match quality is more volatile than stock average match quality and depends on modeling choices such as the number of match-quality states.

11 Figures and tables

11.1 Figure 1. The cleansing and sullyng effects

Read. Figure 1 is the conceptual diagram for the paper. The left part shows the distribution of match surplus before a recession. The middle part shows the effect of a negative productivity shock: some low-surplus matches fall below the destruction threshold and disappear. This is the cleansing effect. The right part adds on-the-job search and shows that the mass of surviving jobs can shift toward lower surplus because fewer workers are moving into better matches.

Interpretation. The figure clarifies why the word “cleansing” can be misleading. It is true that the least efficient matches disappear, but it does not follow that the whole distribution improves. The sullyng effect operates on the surviving part of the distribution. If fewer workers can climb toward high-quality matches, the right tail thins out and the economy accumulates mediocre jobs.

Takeaway. Recessions can simultaneously remove bad matches and reduce the creation or accumulation of very good matches. The net efficiency effect is an empirical and quantitative question, not an implication of job destruction alone.

11.2 Figure 2. Effect of a negative productivity shock on the probability of finding a job

Read. Figure 2 plots the simulated probability $m(v/z, 1)$ of finding a job around a negative productivity shock. The probability is about 0.36 before the shock. It falls sharply to about 0.13 on impact and then slowly recovers to roughly 0.18 after 20 quarters, still far below the prerecession level.

Interpretation. This is the model's direct measure of the sullyng channel. When aggregate productivity falls, firms post fewer vacancies relative to searchers. This makes it harder for workers to move out of mediocre jobs and into better jobs. The probability does recover somewhat because more workers are trapped in low-quality matches and therefore more likely to accept offers, which raises the expected value of a vacancy. But the recovery is incomplete.

Why it matters. The whole quantitative mechanism depends on whether the job-finding probability falls enough in recessions. Figure 2 shows that the calibrated model produces a large fall in the key mobility probability.

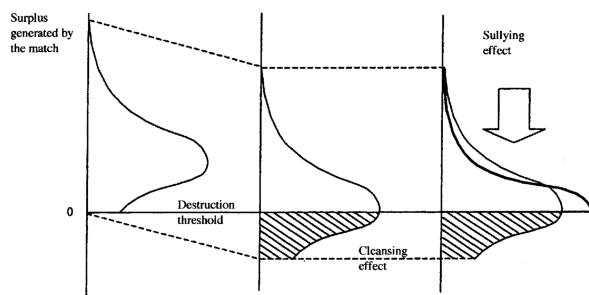


FIGURE 1
The cleansing and sullyng effects

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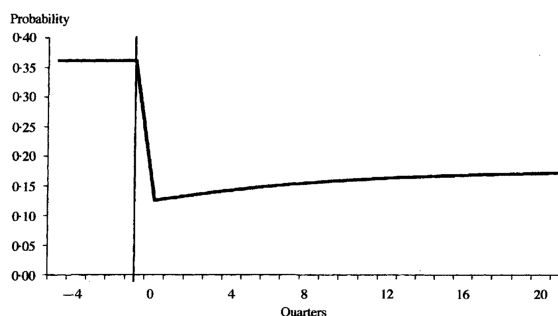


FIGURE 2
Effect of a negative productivity shock on the probability of finding a job

Figure 2. Effect of a negative productivity shock on the probability of finding a job

11.3 Figure 3. Quit rates over the cycle

Read. Figure 3 compares quit rates and unemployment in the data and in the model. The top panel uses PSID quit data and shows that quits are procyclical: they are lower when unemployment is high. The bottom panel reports the same objects in the simulated model, using an aggregate productivity path chosen to mimic U.S. unemployment movements.

Interpretation. The figure provides empirical discipline for the on-the-job search channel. A quit often represents a worker leaving one job for another. If quits fall in recessions, then fewer workers move voluntarily toward better matches. This is exactly the behavior required for sullyng. The model captures the direction of the comovement, although it generates more quit-rate volatility than the data.

Caveat. The paper explicitly notes that the model's very high-frequency quit movements are too volatile. The more relevant comparison is that both the model and the data imply lower job-to-job mobility during downturns.

11.4 Figure 4. Mismatch over the cycle

Read. Figure 4 compares an empirical mismatch proxy with the model's mismatch measure. The top panel uses the fraction of employed PSID workers who report thinking about changing jobs. The bottom panel uses the model's fraction of employed workers with match quality $n < N$. Both are plotted with unemployment.

Interpretation. Mismatch rises when unemployment is high. This is a central piece of evidence against a pure cleansing interpretation. If recessions simply improved matching by removing bad jobs, mismatch should fall. Instead, the figure suggests that recessions leave more workers in jobs they would like to improve upon.

Economic meaning. The model explains countercyclical mismatch as a stock effect. During recessions, less job-to-job mobility prevents workers from climbing to their best matches. Over time, a larger stock of employed workers is trapped in intermediate-quality jobs.

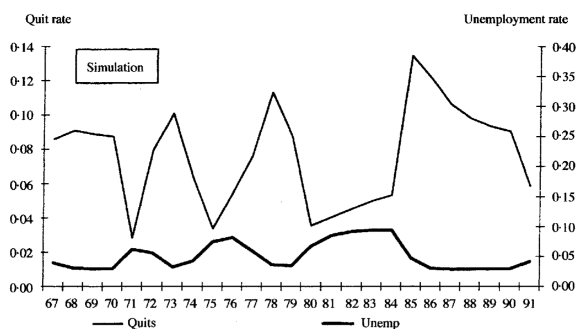
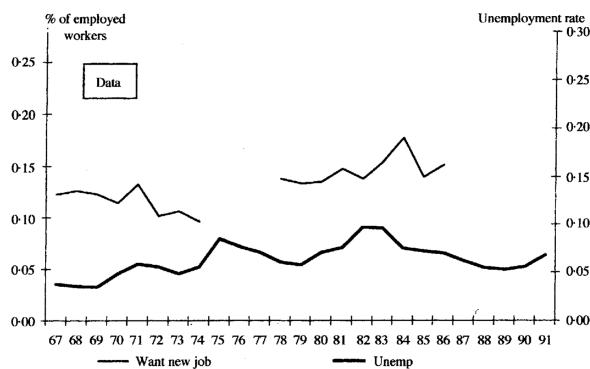
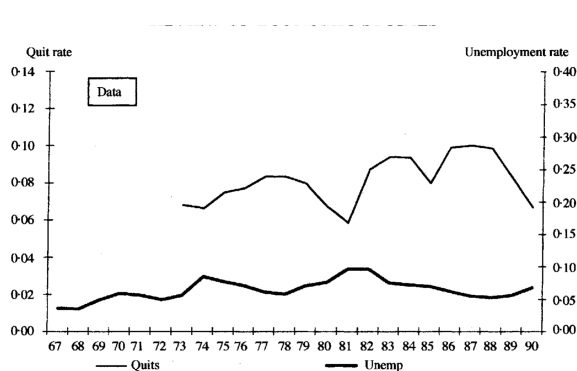


FIGURE 3
Quit rates over the cycle

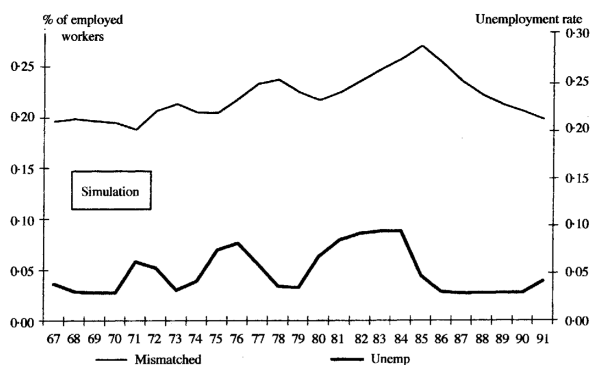


FIGURE 4
Mismatch over the cycle

Figure 3. Quit rates over the cycle

Figure 4. Mismatch over the cycle

11.5 Figure 5. Average match productivity

Read. Figure 5 tracks average match productivity. The top panel shows the response to a one-time negative productivity shock. Average match quality initially increases from about 1.219 to about 1.221 because the worst matches are destroyed. After several quarters, average quality declines below its original level and keeps falling. The bottom panel applies the model to a path that resembles U.S. unemployment over the 1970s and 1980s.

Interpretation. This is the paper’s main quantitative result. Cleansing is visible, but small and short-lived. Sullyng accumulates over time and dominates. The economy gives up the process by which employed workers gradually move toward better jobs, so average match quality declines.

Magnitude. The model predicts a cumulative decline in average match productivity from about 1.219 to about 1.210 by the peak unemployment period, roughly a 1 percent productivity loss. The effect is large enough to matter for aggregate productivity and not just for individual job satisfaction.

11.6 Figure 6. Effect of a negative productivity shock on the quality of new matches

Read. Figure 6 focuses on the average quality of newly created matches after a negative productivity shock. There is a large immediate spike because matches of quality $n = 1$ are no longer created; this is the cleansing effect applied to new matches. But after about five quarters, new-match quality falls below its prerecession value for several years.

Interpretation. The figure shows why jobs created in recessions can look worse in data, even though the very worst matches are screened out. As the stock of workers shifts toward lower-quality current jobs, workers become willing to accept lower-quality new jobs than they would have accepted in a boom. The flow of new matches is therefore temporarily sullied.

Connection to evidence. This mechanism is consistent with evidence that jobs created in recessions are more likely to be temporary, low-paying, or short-lived. The paper cautions that flow measures are more sensitive to parameter choices than stock productivity measures.

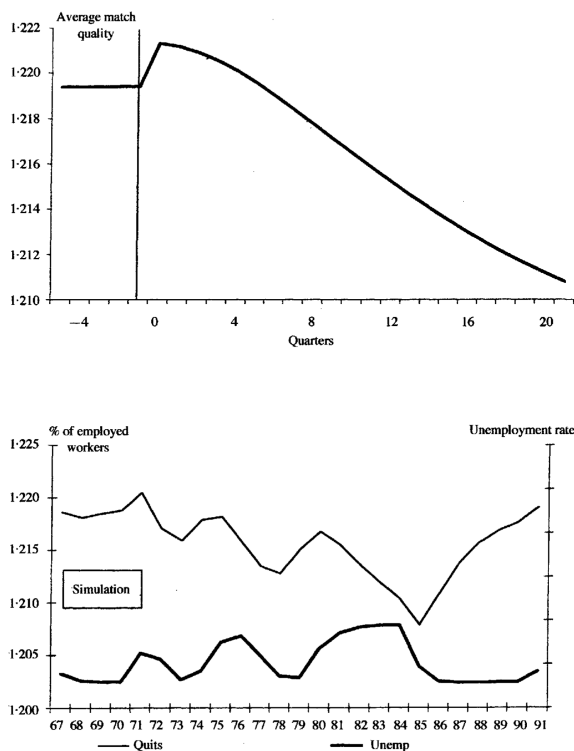


FIGURE 5
Average match productivity

Figure 5. Average match productivity

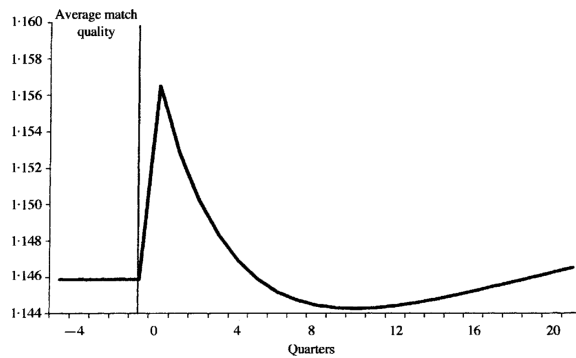


FIGURE 6
Effect of a negative productivity shock on the quality of new matches

Figure 6. Effect of a negative productivity shock on the quality of new matches

12 Bottom line

Barlevy (2002) is best read as a qualification of the Schumpeterian cleansing view. Recessions do remove some inefficient matches, but they also reduce the vacancy creation that allows employed workers to reallocate upward. Once job-to-job mobility is included, recessions may worsen the allocation of labor even while increasing job destruction. The paper's empirical and quantitative exercises suggest that this sullyng effect is economically meaningful and likely larger than the conventional cleansing effect.